

# Eduardo Venegas

Data & AI Engineer

## EDUCATION

**MSc. Data Science and Artificial Intelligence** Universität des Saarlandes 10/2024 - 09/2026 Saarbrücken, Germany

- Coursing. *Statistics* | *ML Cybersecurity* | *NLP* | *Machine Translation* | *Image Processing & Computer Vision*.
- Thesis (Seminar Grade 1.0):** Evaluating Multi-Turn Agentic RAG Systems (In collaboration with BMW & DFKI).

**BSc. Computer Science** Tecnológico de Monterrey GDL

08/2019 - 12/2023 Guadalajara, Mexico

- Grade: 94.6 / 100. Five times best software engineering GPA. **60% tuition** scholarship due to academic merit.
- ML* | *Time series* | *Algorithms & Optimization* | *Cloud Computing & Distributed* | *Vehicle User Interfaces* | *PLC*.

**Computer Science Exchange Student** Universität Ulm

10/2022 - 09/2023 Ulm, Germany

- Full scholarship** recipient from **DAAD**. *Computer Vision II* | *Autonomous Driving and Assistance Systems* | *Statistics*.

**Technologist Degree in Automation and Robotics** CETI Colomos

08/2015 - 06/2019 Guadalajara, Mexico

- Control Laboratory (I & II)* | *Industrial Robotics* | *PLCs* | *Digital Electronics* | *Controllers Laboratory* | *Sensor Systems*

## SKILLS

### Data

RAG LangChain LangGraph HuggingFace MCP R SQL TensorFlow Knowledge Graphs

### Software

JavaScript TypeScript Python C C++ React Angular Flask Node Ionic/Tauri Matlab

### Systems

Linux Docker Kubernetes AWS CI/CD Grafana MQTT PLC Arduino

## WORK EXPERIENCE

**Data Scientist Masterarbeit** BMW & Deutsches Forschungszentrum für KI (DFKI)

03/2026 - 09/2026 Munich, Germany

- Topic:** *Evaluating Multi-Turn Agentic RAG Systems: A Synthetic Benchmark Generation Approach*.
- Architected a modular evaluation framework for **tool-using Agentic RAG** systems to enable fine-grained failure localization.
- Built a **multi-turn synthetic benchmark** that combines deterministic **task planning** with diverse user **personas**.
- Engineered an **unsupervised retrieval engine** to audit a **RAG system** over **4M unannotated** corporate records.

**Data Scientist Werkstudent** August-Wilhelm Scheer Institute

10/2024 - 02/2026 Saarbrücken, Germany

- Developed a **KG-RAG** over unstructured technical documentation and a **Planning Agent**, using **LangChain** and **HuggingFace**.
- Implemented an **Autoencoder** model for **Anomaly Detection** via **Federated Learning** across edge devices.

**Cloud Software Engineer** Oracle

11/2023 - 10/2024 Guadalajara, Mexico

- Worked on the **Oracle Cloud Infrastructure** in the team of **Big Data Service** in **Oracle Analytics**.
- Developed tools for managing and deploying **big data clusters**. Implemented **pipelines** for automation and testing.

**Data Analyst Intern** BMW Group

03/2023 - 09/2023 Munich, Germany

- Developed a high-accuracy algorithm for **detecting external battery replacements**.
- Identified **root cause** of **missing db values** and created a new processing pipeline, **increasing service demand by 62%**.
- Designed a **Streamlit dashboard** for adjusting prediction model parameters and obtaining a **real-time output (<8 seconds)**.
- Configured **cloud-based environments** for working with **PySpark**.

**Site Reliability Engineer** Xira

01/2022 - 04/2022 Mexico City, Mexico

- Managed** over 10 servers, **monitored** services and **enhanced** engineering processes.
- Built a **platform** to manage client **APIs**, integrating payments and subscription systems using **NextJS**, **Stripe** and **Docker**.

**Production Engineer Fellow** Major League Hacking & Meta

06/2021 - 09/2021 New York, USA

- Selected among 120 out of 20,000 applicants. Implemented monitoring with **Prometheus** and **Grafana**.
- Automated **CI/CD** workflow for efficient deployment to a production server using **Docker** in an **AWS EC2** instance

**Full Stack & Data Engineer** Jiit Technologies

10/2020 - 06/2021 Guadalajara, Mexico

- Designed a custom sales data analysis pipeline API using **Flask**, **SciPy** and **Prophet**.
- Built an **ERP** platform from scratch to manage inventory, human resources, shipments, income and expenses. **Angular**.

## PROJECTS

**VectorDrive-Route: Autonomous BEV Trajectory Planning**

PyTorch ResNet Computer Vision Github

- Built a **PyTorch** pipeline using **nuScenes**, **ResNet** and **depth lifting** to map multi-camera feeds into a **Bird's-Eye-View** grid.
- Designed a **temporal motion planning head** with **drivable-area** and **occupancy decoders** to generate safe paths.

**Daten-Mapping-Framework für die AAS**

AAS Industry 4.0 LLM DOI: 10.17560/atp.v68i5.2836

- Developed a mapping **framework** from schemas to **Asset Administration Shell** to achieve interoperability in **Industry 4.0**.
- Designed a dual alignment strategy separating standard **relational** tables from **hierarchical** assets.

**See Your World**

Tensorflow Google Cloud Flask Github

- Crafted a hackathon **award-winning** app for the **visually impaired**, supporting **11 languages**.
- Engineered **Image Captioning** DL model with **TensorFlow**, deployed using **Flask**, **Docker**, and **Google Cloud VM**.

## CERTIFICATIONS

- Mathematics for Data Science Specialization
- Data Science Professional Certificate
- Python Data Science Professional Certificate
- Deep Learning Professional Certificate

Higher School of Economics - 2021

Harvard - 2020

IBM - 2020

IBM - 2020

## LANGUAGES

Spanish Native ●●●●●●

English C1 ●●●●●●

German C1 ●●●●●○

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